

ICT 4442 Deep Learning Mini Project Guidelines

1. Overview

Each team applies “Deep Learning” to a real-world problem of their choice (within the scope of the course), implementing and comparing multiple model architectures. The project is evaluated across three phases: Synopsis, Interim, and Final Presentation, for a total of 10 marks.

2. Team Formation

- Maximum group size: 4 members max.
- Each team must select 3–4 models to implement and compare, ideally one model (or model family) per member, so each member's contribution is independently demonstrable.

3. Phases, Deadlines & Deliverables

Phase	Deadline	Deliverable	Weight	Marks
Phase 1 Synopsis Submission	Tue, 25 Aug 2026	Report as per Report Template Part A, containing: • Team No. & Names, Title, Brief description (≤ 500 words) • Dataset(s) identified (with details of the dataset) • 3–4 proposed models for comparison (spanning different architecture families) + model-to-member assignment • Existing state-of-the-art method(s) related to the project • Objectives (crisp, 2–3) & importance/motivation of the project • Preliminary reference list (minimum 5 papers) • Work plan table (milestones with expected completion dates)	10%	1
Phase 2 Interim Submission	Fri, 25 Sept 2026 (before Mid-term Exam on 1 Oct)	Report as per Report Template Part B, containing: • Completed literature review: 8–10 papers in a summary table (paper / method / dataset / key result / relevance) • Dataset acquisition & preprocessing pipeline (code + description) • At least 2 of the 3–4 proposed models implemented with preliminary (even unoptimized) results • Individual contribution log to date (commit history + task matrix signed by all members) • Risk/plan for remaining models and timeline to Final submission	30%	3
Phase 3 Final Presentation & Report	Presentations: 26–29 Oct 2026 Report + code: Sat, 31 Oct 2026 (hard deadline)	Report as per Report Template Part C + code repo, containing: • Full report: Abstract, Keywords, Introduction, Literature Review, Objectives, Methodology (one subsection per model, written by that model's owner), Experimental Setup, Results & Discussion (comparison table across all 3–4 models on common	60%	6

Phase	Deadline	Deliverable	Weight	Marks
		metrics), Conclusion, Future Work, Acknowledgement, References (12+) • Individual Contribution Statement (Part C.3, mandatory extra page, signed by all members) • In-class presentation (10–12 min) + Q&A		

4. Marks Distribution & Rubric

Criterion	Marks	Notes
Synopsis: problem clarity, feasibility, model selection rationale, initial literature	1.0	All-or-nothing on timely, complete submission; resubmission not allowed
Interim: literature review depth (8–10 papers)	1.0	Table format; must include comparison, not just listing
Interim: preprocessing + ≥ 2 models with preliminary results	1.5	Evidence in repo (notebooks/scripts + commit history)
Interim: individual contribution evidence to date	0.5	Verified via GitHub commits / task log
Final: model implementation & comparative performance (3–4 models, common metrics)	2.5	Must include a results/comparison table and error analysis
Final: individual contribution, each member defends their assigned model in viva	1.5	Per-member score can differ from the group score by up to $\pm 20\%$ based on viva
Final: report quality & integrated literature review	1.0	Structure, clarity, correct citations
Final: presentation & Q&A handling	1.0	All members must present part of the work

5. Model Comparison Requirement

- Minimum 3, maximum 4 models, chosen to span different architecture families covered in the course (e.g., a classical/MLP baseline, a CNN variant, an RNN/LSTM variant, or an attention-based model), so the comparison is meaningful and each member owns a distinct piece.
- Report a common evaluation protocol: same train/val/test split, same metrics (accuracy, precision/recall/F1, IoU, BLEU, etc. as appropriate to the task), and a single comparison table.
- Discuss why performance differs across models.

6. Literature Review Requirement

- Interim: minimum 8–10 papers, summarized in a table (paper, method, dataset, key result, relevance to the project).
- Final: literature review integrated into the report's Related Work section, framing how the chosen models relate to prior work.

7. Individual Contribution Policy

- Each team must submit a signed Individual Contribution Statement at Interim and Final stages, listing who did what (data prep, specific model implementation, tuning, evaluation, writing, slides).
- GitHub commit history is used to cross-check contribution claims, teams should use a shared repository with per-member commits, not a single push at the end.
- At the Final Presentation, each member is questioned individually on the model they implemented. Marks for the Final phase (6 marks) may be adjusted per member by up to $\pm 20\%$ of the group score based on individual viva performance.
- A member who cannot explain their assigned model in the viva will not receive full marks for that component, regardless of the group's overall result.

8. Submission Format

- Synopsis: Report as per the Report Template (Part A), submitted via MS Forms by 11:59 PM on the deadline date.
- Interim: Report as per the Report Template (Part B), submitted via MS Forms by 11:59 PM on the deadline date. (No GitHub repo link required at this stage.)
- Final: Report as per the Report Template (Part C) + GitHub repo link, submitted via MS Forms by 11:59 PM, 31 Oct 2026.
- Late submissions: appropriate marks will deducted and report not accepted after that without prior approval by the faculty.

9. Academic Integrity

- Code and text must be original; any use of external code/tutorials/LLMs must be cited in the report and mention where exactly LLM was used.
- Plagiarism or fabricated results in any phase will result in zero marks for that phase for the involved member(s). Therefore, the students must produce a Turnitin plagiarism report and %AI content with $\leq 15\%$ of plagiarism and $\leq 20\%$ AI content acceptable.